```
ishaan_dhar@Ishaan:~$ ping -c 5 92.4.73.213
PING 92.4.73.213 (92.4.73.213) 56(84) bytes of data.
64 bytes from 92.4.73.213: icmp_seq=1 ttl=57 time=66.0 ms
64 bytes from 92.4.73.213: icmp_seq=2 ttl=57 time=65.5 ms
64 bytes from 92.4.73.213: icmp_seq=3 ttl=57 time=64.5 ms
64 bytes from 92.4.73.213: icmp_seq=4 ttl=57 time=68.3 ms
64 bytes from 92.4.73.213: icmp_seq=5 ttl=57 time=65.0 ms

--- 92.4.73.213 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 64.476/65.840/68.274/1.312 ms
```

Here, there is a TTL of 57. It is lower in this case as the VM runs Ubuntu, and Linux has a default assigned TTL of 64. So, doing the math, I find that the number of hops undertaken by the packets to reach the destination from the source comes to 7 hops (64-57=7). This is easy to understand, because instead of a 'distant' google server, my request is being transferred straight to Oracle's Mumbai server, which needs the packets to not go through as many routers to reach the destination on the way back. The average RTT IS 65.840 which is significantly higher than the RTT for pinging google.com, as in this case, the traffic has to go through multiple channels, instead of straightaway going to google.com through direct connections as in that case. Here, the traffic has to go through the ISP and its channels before actually being routed to the target server.

Below, the commands that could have been run have been shown.

```
ishaan_dhar@Ishaan:~$ nslookup 92.4.73.213
** server can't find 213.73.4.92.in-addr.arpa: NXDOMAIN

ishaan_dhar@Ishaan:~$ nslookup google.com
Server:         10.255.255.254
Address:        10.255.255.254#53

Non-authoritative answer:
Name:   google.com
Address: 172.217.24.206
Name:   google.com
Address: 2404:6800:4007:805::200e

ishaan_dhar@Ishaan:~$ nslookup google.com 1.1.1.1
Server:         1.1.1.1
Address:        1.1.1.1#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.205.110
Name:   google.com
Address: 2404:6800:4007:832::200e
```

There is no actual DNS server for the VM, so it shows a can't find message.

```
ishaan_dhar@Ishaan:~$ traceroute 92.4.73.213
traceroute to 92.4.73.213 (92.4.73.213), 30 hops max, 60 byte packets
 1  172.30.160.1 (172.30.160.1)  0.962 ms  0.832 ms  0.792 ms
 2  dsldevice.lan (192.168.6.1)  9.675 ms  9.558 ms  9.525 ms
 3  abts-kk-dynamic-001.4.179.122.airtelbroadband.in (122.179.4.1)  9.418 ms  9.389 ms  9.057 ms
 4  125.18.238.241 (125.18.238.241)  8.952 ms aes-static-169.82.22.125.airtel.in (125.22.82.169)  8.923 ms 125.21.0.229 (125.21.0.229)  12.549 ms
 5  182.79.240.3 (182.79.240.3)  29.444 ms 182.79.240.1 (182.79.240.1)  51.597 ms 182.79.211.35 (182.79.211.35)  38.288 ms
 6  125.17.12.178 (125.17.12.178)  70.511 ms  41.452 ms  36.702 ms
 7  140.91.204.10 (140.91.204.10)  36.623 ms 140.91.204.21 (140.91.204.21)  35.853 ms 140.91.204.23 (140.91.204.23)  41.665 ms
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
```

The route here probably shows the packet reaching the Oracle server, which then leads to unresponsive * * * messages, likely due to Oracle not allowing UDP traffic inside the server, which is a common method.

```
ishaan_dhar@Ishaan:~$ sudo traceroute -T -p 22 92.4.73.213
traceroute to 92.4.73.213 (92.4.73.213), 30 hops max, 60 byte packets
 1  172.30.160.1 (172.30.160.1)  1.237 ms  10014.925 ms *
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  92.4.73.213 (92.4.73.213)  28.940 ms  42.200 ms  42.129 ms
```

Here, I am sending TCP SYN packets to port 22, which means it looks like an ssh request to the server, which then allows the traffic through. Here, it shows that on the 8th hop it reaches the target, which corresponds with my previously calculated hop count of 7.

```
ishaan_dhar@Ishaan:~$ curl https://92.4.73.213
 ^C
ishaan_dhar@Ishaan:~$ curl https://92.4.73.213
^C
ishaan_dhar@Ishaan:~$ curl -v telnet://92.4.73.213:22
*   Trying 92.4.73.213:22...
* Established connection to 92.4.73.213 (92.4.73.213 port 22) from 172.30.163.219 port 38758
SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.16
^C
```

Here, there is no response from the server because there is no SSL certificate or web server 'listening' for https connections on port 443, which is the port number for https. I additionally used the telnet feature to force a SYN, SYN-ACK, ACK (TCP 3 way handshake), which caused my VM to send me the OpenSSH version string before the connection just hanged.